

Tópicos 3 e 4

Medidas de Posição

- Média	Popl μ	Amostra $\bar{x}$
- Mediana	$MD(x)$	$mo(x)$
- Moda	$Mo(x)$	$mo(x)$

Notações

Pop.: $X_1, X_2, \dots, X_N$; N elementos $n \leq N$

Amostra: $X_1, X_2, \dots, X_n$; n elementos

Média

Pop. $X_1, X_2, \dots, X_N$

$$\mu = \sum_{i=1}^N X_i / N = (X_1 + X_2 + \dots + X_N) / N$$

Amostra

$$\bar{x} = \sum_{i=1}^n X_i / n = (X_1 + X_2 + \dots + X_n) / n$$

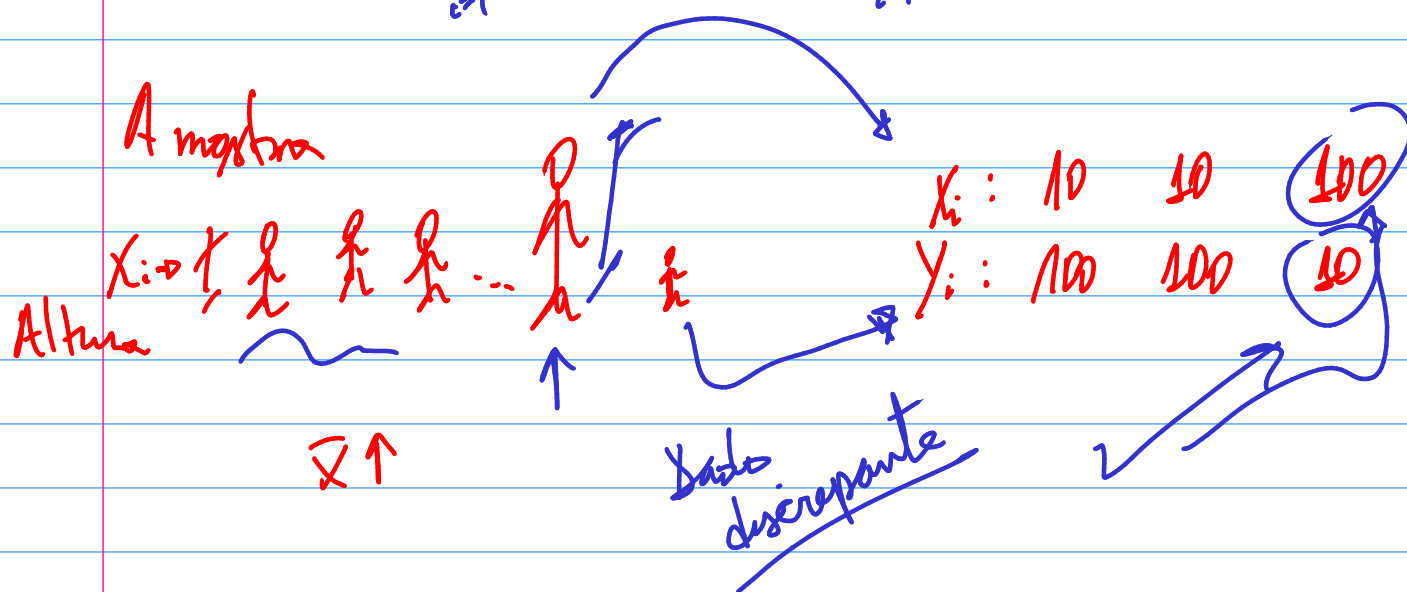
Nas
Amostras

Ex:

					$\tilde{x}_i$							
			Classes	F_i	PM	Fr	Fac1	Fac2	Fp	Fac1p	Fac2p	
$k=$	1	95.7	---	97.01	2	96.355	0.04	2	50	4	4	100
$k=$	2	97.01	---	98.32	7	97.665	0.14	9	48	14	18	96
	3	98.32	---	99.63	14	98.975	0.28	23	41	28	46	82
	4	99.63	---	100.94	13	100.285	0.26	36	27	26	72	54
	5	100.94	---	102.25	9	101.595	0.18	45	14	18	90	28
	6	102.25	---	103.56	3	102.905	0.06	48	5	6	96	10
	7	103.56	---	104.87	2	104.215	0.04	50	2	4	100	4

Pop. $\mu = \frac{\sum_{i=1}^k \tilde{x}_i f_i}{N} ; N = \sum_{i=1}^k f_i$

Amost. $\bar{x} = \frac{\sum_{i=1}^k \tilde{x}_i f_i}{n} ; n = \sum_{i=1}^k f_i$



Mediana

Pop.: $N=3$

$X_{(1)} \quad X_{(2)} \quad X_{(3)} \quad X_{(4)}$

$X_1 \quad X_2 \quad X_3 \Rightarrow X_{(1)} \quad X_{(2)} \quad X_{(3)}$

N é ímpar

$$Md(X) = \begin{cases} X_{(\frac{N+1}{2})} & , \text{ se } N \text{ é ímpar} \\ \frac{X_{(\frac{N}{2})} + X_{(\frac{N}{2}+1)}}{2} & , \text{ se } N \text{ é par} \end{cases}$$

Amostragem:

$$Md(X) = \begin{cases} X_{(\frac{n+1}{2})}, & \text{se } n \text{ é ímpar} \\ \frac{X_{(\frac{n}{2})} + X_{(\frac{n}{2}+1)}}{2}, & \text{se } n \text{ é par} \end{cases}$$

Para dados agrupados:

$$Md(X) = Lt + h \frac{\frac{n}{2} - f_{ant}}{f_{md}} \cdot C$$

100,94 - 99,63

Classe de
Mediana $\Rightarrow \frac{n}{2}$
 $n=50 \Rightarrow 25$

	Classes	Fi	PM	Fr	Fac1	Fac2	Fp	Fac1p	Fac2p
1	95.7 ---	97.01	2	96.355	0.04	2	50	4	100
2	97.01 ---	98.32	7	97.665	0.14	9	48	14	96
3	98.32 ---	99.63	14	98.975	0.28	23	41	28	82
4	99.63 ---	100.94	13	100.285	0.26	36	27	26	72
5	100.94 ---	102.25	9	101.595	0.18	45	14	18	90
6	102.25 ---	103.56	3	102.905	0.06	48	5	6	96
7	103.56 ---	104.87	2	104.215	0.04	50	2	4	100

Moda

DA $Mo(X) = \text{Valor mais freq.}$

$$\Delta_1 = 14 - 7$$

$$\Delta_2 = 14 - 13$$

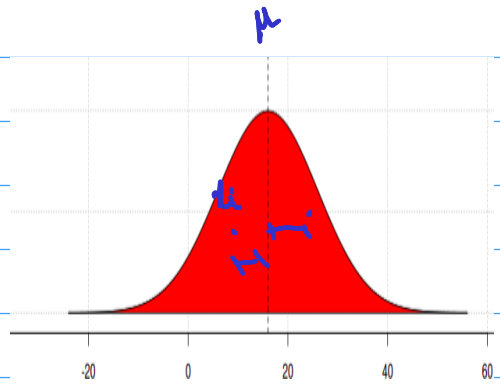
$$DA \quad Mo(X) = Lt + h \frac{\Delta_1}{\Delta_1 + \Delta_2} \cdot C$$

Classe da
Moda

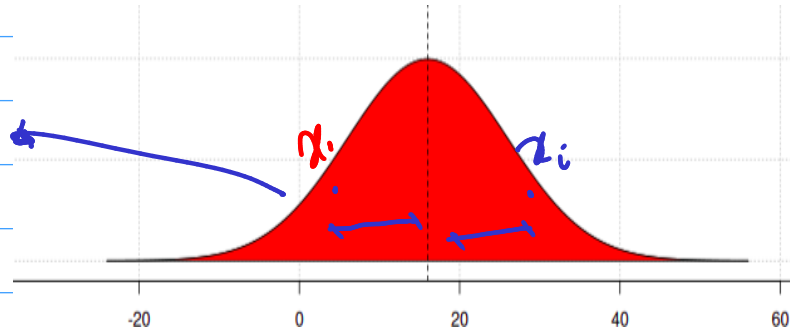
	Classes	Fi	PM	Fr	Fac1	Fac2	Fp	Fac1p	Fac2p
1	95.7 ---	97.01	2	96.355	0.04	2	50	4	100
2	97.01 ---	98.32	7	97.665	0.14	9	48	14	96
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Medidas de Variabilidade

- Variância
- Desvio Padrão
- Coef. de Variação



$$\sum_{i=1}^N (x_i - \mu) = 0$$



$$(x_1 - \mu) + \dots + (x_N - \mu)$$

$$[(x_1 + \dots + x_N) - N\mu]$$

$$\mu = \frac{x_1 + x_2 + \dots + x_N}{N}$$

$$SQ = \sum_{i=1}^N (x_i - \mu)^2$$

$$SQ = \sum_{i=1}^N (x_i - A)^2$$

$\frac{dSQ}{dA} \Rightarrow A = \mu$

Variância

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Amplitude

$$A = x_{(N)} - x_{(1)}$$

Coef. Variação

Desvio Padrão

$$\sigma = \sqrt{\sigma^2}$$

G_1
 $\sigma = 4$
 $\bar{x} = 10$

G_2
 $\sigma = 2$
 $\bar{x} = 4$

Coef. de Variación (CV)

$CV_1 = 40\%$ $CV_2 = 50\%$

$$CV = \frac{\sigma}{\mu} \times 100$$

Unidades de X

	σ^2	σ	CV
Unid. X cm	cm ²	cm	Adm